

Modeling

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Solution of the practice quiz

The following binary decision variables are equal to 1 if the corresponding candidate is hired, and 0 otherwise:

- x_1 : Asmodeus,
- x_2 : Belphegor,
- x_3 : Beelzebub,
- x_4 : Leviathan,
- x_5 : Lucifer,
- x_6 : Mammon,
- x_7 : Satan.

The constraints are modeled as follows.

1. Not all candidates can be hired. It means that not all variables can take the value one. We require the sum not to exceed 6:

$$\sum_{i=1}^7 x_i \leq 6.$$

2. At least one candidate must be hired. At least one variable must take the value one. We require the sum to be at least equal to one.

$$\sum_{i=1}^7 x_i \geq 1.$$

3. Asmodeus can not be hired if Beelzebub is. The fact that Asmodeus is not hired is represented by a negation: $1 - x_1$. The implication is then modeled as

$$x_3 \leq 1 - x_1,$$

or, equivalently,

$$x_1 + x_3 \leq 1.$$

4. Leviathan can be hired only if Belphegor is. If $x_4 = 1$, then $x_2 = 1$. It is modeled as

$$x_4 \leq x_2,$$

or equivalently,

$$x_2 - x_4 \geq 0.$$

5. Asmodeus and Lucifer are not separable, and have to be hired both, or not at all. Basically, it means that the decision for one is the same as the decision for the other one. The constraint is

$$x_1 = x_5,$$

or, equivalently,

$$x_1 - x_5 = 0.$$

6. One of the two following constraints must be verified. Either at least one of the candidates in the group Asmodeus, Belphegor, Beelzebub is chosen, or at least two in the group Belphegor, Leviathan, Lucifer, Mammon. To model this, we define one additional binary variables y . y is 1 if the constraint on the first group applies, and 0 otherwise. Now, if the constraint applies to group 1, that is, if $y = 1$, we have

$$x_1 + x_2 + x_3 \geq y = 1,$$

or, equivalently,

$$x_1 + x_2 + x_3 - y \geq 0.$$

Note that, if $y = 0$, the constraint is verified for any combination of values of the binary variables. Similarly, if the constraint applies to group 2, that is, if $y = 0$, we have

$$x_2 + x_4 + x_5 + x_6 \geq 2(1 - y) = 2,$$

or, equivalently,

$$x_2 + x_4 + x_5 + x_6 + 2y - 2 \geq 0.$$

Note that, if $y = 1$, the constraint is verified for any combination of values of the binary variables. Note also that it is allowed for the two constraints to be verified. This is illustrated in the feasible solution provided below.

One feasible solution is

$$\begin{array}{ll} x_1 = 0, & x_2 = 1, \\ x_3 = 1, & x_4 = 1, \\ x_5 = 0, & x_6 = 1, \\ x_7 = 1, & y = 1. \end{array}$$

It consists in hiring Belphegor, Beelzebub, Leviathan, Mammon, and Satan. We verify the constraints:

$$\begin{aligned} \sum_{i=1}^7 x_i &= 5 \leq 6, \\ \sum_{i=1}^7 x_i &= 5 \geq 1, \\ x_1 + x_3 &= 1 \leq 1, \\ x_2 - x_4 &= 0 \geq 0, \\ x_1 - x_5 &= 0, \\ x_1 + x_2 + x_3 - y &= 1 \geq 0, \\ x_2 + x_4 + x_5 + x_6 + 2y - 2 &= 3 \geq 0. \end{aligned}$$